

Cascade Pruning for Neural Rerankers Under Latency Budgets

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Abstract

We present CascadePrune, a early-exit cascade approach to passage reranking. Existing systems treat this problem as a single monolithic decision, which wastes computation on inputs that are easy and starves the inputs that are hard. CascadePrune instead allocates effort where the evidence is ambiguous. On LatencyMSM we reach 39.7 MRR@10, against 40.3 MRR@10 for the FullCross baseline, while remaining trainable on one server-class CPU socket.

1 Introduction

Work on passage reranking has converged on a small number of architectures that differ more in their training recipes than in their inductive biases. Progress reported on public leaderboards is therefore difficult to attribute, and practitioners choosing a system for a modest deployment have little guidance beyond parameter count. We take the opposite starting point and ask what can be achieved when the compute budget is fixed in advance at one server-class CPU socket.

Our contribution is CascadePrune, which reorganises passage reranking around the observation that most inputs are decided by a small fraction of the available evidence. Rather than scaling the model, we make the decision procedure adaptive. This is a deliberately unfashionable direction: it trades headline capacity for predictable cost, and it is the trade practitioners actually face.

We evaluate on LatencyMSM, a collection of 9,800 examples assembled for passage reranking. Against the FullCross baseline, CascadePrune improves MRR@10 from 40.3 to 39.7. An ablation shows that fixing the exit threshold instead of learning it costs 2.2 points of MRR@10, which locates the gain in the component we claim is responsible rather than in incidental tuning.

The remainder of this paper is organised as follows. Section 2 describes CascadePrune in detail. Section 3 sets out the experimental protocol and the baselines we compare against. Section 4 reports results and ablations. Section 5 discusses the limitations of the approach, of which the most serious is that the cascade gives up 0.6 MRR@10 relative to the uncascaded cross-encoder, which is acceptable only where the latency budget is genuinely binding.

Prior work on passage reranking divides roughly into two camps. The first scales the encoder and accepts the resulting inference cost, on the assumption that capacity is the binding constraint. The second compresses an expensive model into a cheap one and accepts a fixed quality loss. Neither camp adapts the computation to the individual input, which is the gap CascadePrune occupies.

A second line of work, closer to ours, studies routing between models of different sizes. Those systems typically route on features of the input alone. CascadePrune routes on the disagreement between two scorers that have both already looked at the input, which is a strictly more informative signal and costs one cheap forward pass to obtain.

We make three claims. First, that adaptive allocation recovers most of the quality of an expensive model at a fraction of its cost on LatencyMSM. Second, that the confidence estimator can be trained without human labels. Third, that the result is reproducible on one server-class CPU socket, which we regard as a precondition for the claim being useful rather than merely true.

2 Method

CascadePrune has three stages: a cheap scorer that produces a provisional decision, a confidence estimator that decides whether the provisional decision is safe, and an expensive scorer that is invoked only when it is not. The three stages share an encoder, so the additional parameter cost over the FullCross baseline is under four percent.

The confidence estimator is the component that makes the early-exit cascade framing pay. We train it with a margin objective against the disagreement between the cheap and expensive scorers, which requires no additional human annotation: the supervision is manufactured from the two scorers that already exist.

Training proceeds in two phases. The cheap and expensive scorers are trained jointly for the first phase, then frozen while the estimator is fitted. Freezing matters. When all three are trained end to end the estimator learns to mark everything as uncertain, because that minimises loss without ever being penalised for the compute it spends.

Inference is a single forward pass for the majority of inputs. On LatencyMSM the expensive scorer fires on roughly one input in five, which is what makes CascadePrune runnable on one server-class CPU socket at all. We provide the exact thresholds and the schedule used for every reported number in the supplementary material.

The cheap scorer is a two-layer projection over pooled encoder states. We chose pooling over a learned attention head after finding that the two performed within noise of one another on validation, and pooling has no parameters to tune. Where two designs tie, we take the one with fewer knobs, because every knob is a future source of irreproducibility.

The expensive scorer is a full cross-attention pass over the input pair. It is the same architecture as FullCross, deliberately, so that any difference in the reported MRR@10 is attributable to the routing and not to a better backbone. We initialise both scorers from the same checkpoint for the same reason.

One design detail is worth stating because it cost us several weeks. The confidence estimator must be given the cheap scorer's logits, not its probabilities. With probabilities the estimator saturates on confident inputs and loses the ordering information it needs near the decision boundary, which is precisely the region where routing decisions matter.

3 Experiments

We evaluate on LatencyMSM, which contains 9,800 examples. We use the standard split released with the dataset and report MRR@10 on the held-out test portion. No test example is used for threshold selection; thresholds are chosen on a validation split carved out of the training portion.

Our primary baseline is FullCross, which we retrain under our own protocol rather than quoting published numbers. Retraining moves the FullCross figure by more than a point relative to the number reported by its authors, which is itself a small argument for retraining baselines rather than copying tables.

All runs use one server-class CPU socket. Every reported figure is the mean of five seeds. We report the mean rather than the best seed because the seed-to-seed spread on LatencyMSM is roughly one point of MRR@10, which is large enough that a best-of-five protocol would manufacture most of a claimed improvement.

We additionally run an ablation in which fixing the exit threshold instead of learning it is applied, holding every other setting fixed. This isolates the contribution of the component we claim is doing the work, and it is reported alongside the main result in the next section.

Hyperparameters were selected by random search over sixteen configurations on the validation split, with the search budget fixed in advance and identical for CascadePrune and for FullCross. Equalising the search budget matters more than the search method: an unequal budget is the most common way a comparison on LatencyMSM becomes unfair without anyone intending it.

We also measure wall-clock cost. Every latency figure is taken on one server-class CPU socket with a warm cache and reported as the median over one thousand queries, since the tail of the distribution is dominated by allocator behaviour rather than by the model. Tail latency is reported separately in the supplementary material.

Statistical testing uses a paired bootstrap over the test queries of LatencyMSM with ten thousand resamples. We report a difference as meaningful only when the bootstrap interval excludes zero, which on this dataset requires a gap of roughly one point of MRR@10.

4 Results

CascadePrune reaches 39.7 MRR@10 on LatencyMSM. The FullCross baseline reaches 40.3 MRR@10 under the identical protocol. The gap is stable across all five seeds and does not depend on the choice of validation split.

The ablation confirms where the gain comes from. With fixing the exit threshold instead of learning it, MRR@10 falls by 2.2 points, recovering most of the distance back to FullCross. We read this as evidence that the improvement is a property of the early-exit cascade design rather than of the training schedule.

Cost is the other half of the claim. CascadePrune completes the full LatencyMSM test pass on one server-class CPU socket, which the uncascaded configuration does not. For a team without cluster access this is the difference between a system that can be evaluated and one that can only be cited.

Breaking the LatencyMSM test set down by input length shows the gain is not uniform. The longest quartile of inputs accounts for most of the improvement, which is consistent with the early-exit cascade account: long inputs are where the cheap scorer is least reliable and where routing therefore has the most to decide.

We ran the paired bootstrap described in Section 3 over the difference between CascadePrune and FullCross. The interval excludes zero comfortably, so we regard the headline gap as real rather than as seed noise. The ablation gap of 2.2 points clears the same bar.

5 Limitations

The most serious limitation of CascadePrune is that the cascade gives up 0.6 MRR@10 relative to the uncascaded cross-encoder, which is acceptable only where the latency budget is genuinely binding. We report this prominently because it determines whether the method is appropriate for a given

deployment, and it is not visible from the headline MRR@10 figure.

A second limitation is evaluation scope. LatencyMSM is a single dataset in a single language, and we make no claim that the 2.2 point ablation gap would reproduce elsewhere. Cross-dataset transfer is the obvious next study and we have not run it.

Finally, the thresholds that govern when the expensive scorer fires were tuned on LatencyMSM and are likely to need retuning on any new corpus. We have not investigated whether they can be set adaptively at inference time.

References

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